Homework — 1.4.3 Boolean Algebra

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 1.4.3 lesson. SHOW ALL WORKING.

Part A — This week's topic (~14 marks)

1. Complete the truth table for the expression **Q = (A + B) · ¬C**. **[4]** *(AO2)*

A	B	C	A+B	¬C	Q
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

2. Use **De Morgan's law** (and double negation where needed) to simplify `¬(A · ¬B)` to an expression with no outer bar. Show each step. **[3]** *(AO2)*

Working space:

3. Simplify the expression `A + ¬A · B` as far as possible, and name the law(s) used. **[3]** *(AO1/AO2)*

Working space:

4. State the Boolean expressions for the **Sum** and **Carry** outputs of a **half adder**, and explain why a half adder **cannot** be used on its own to add the middle columns of a multi-bit binary addition. **[2]** *(AO1)*

Working space:

5. State **two** differences between **combinational** logic and **sequential** logic, giving one example component of each. **[2]** *(AO1)*

Working space:

Part B — Retrieval: keep it fresh (~6 marks)

6. Explain the difference between **symmetric** and **asymmetric** encryption, and state **one** advantage of asymmetric encryption. *(1.3)* **[3]** *(AO1)*

Working space:

7. Describe how the **round robin** CPU scheduling algorithm works, and state **one** advantage of it. *(1.2)* **[3]** *(AO1)*

Working space: